

Quiz 1A

**Q1-01-A**

Give an algorithm to compute  $x^{77}$  that only uses multiplications and uses as few multiplications as possible. How many multiplications does your algorithm use?

**Q1-02-A**

Convert from base 10 to base 2: (a)  $(14.75)_{10}$ , (b)  $(2.7)_{10}$ .

**Q1-03-A**

Convert from base 2 to base 10: (a)  $(11110.01)_2$ , (b)  $(1.0011)_2 \times 2^5$ .